



- 8 A polished sheet of magnesium in a vacuum emits electrons when it is illuminated by ultraviolet radiation.

(a) State the name of this phenomenon.

..... [1]

- (b) For emission of electrons to occur, the frequency of the ultraviolet radiation must be at least 8.8×10^{14} Hz.

(i) Calculate the work function energy of magnesium.

work function energy = J [2]

- (ii) For ultraviolet radiation with a frequency of 11×10^{14} Hz, calculate the maximum speed of the emitted electrons.

maximum speed = ms^{-1} [3]



- (c) The frequency f of the ultraviolet radiation incident on the magnesium sheet is varied between $8.0 \times 10^{14} \text{ Hz}$ and $11 \times 10^{14} \text{ Hz}$.

On Fig. 8.1, sketch the variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons. Use the space below for any working that you need.

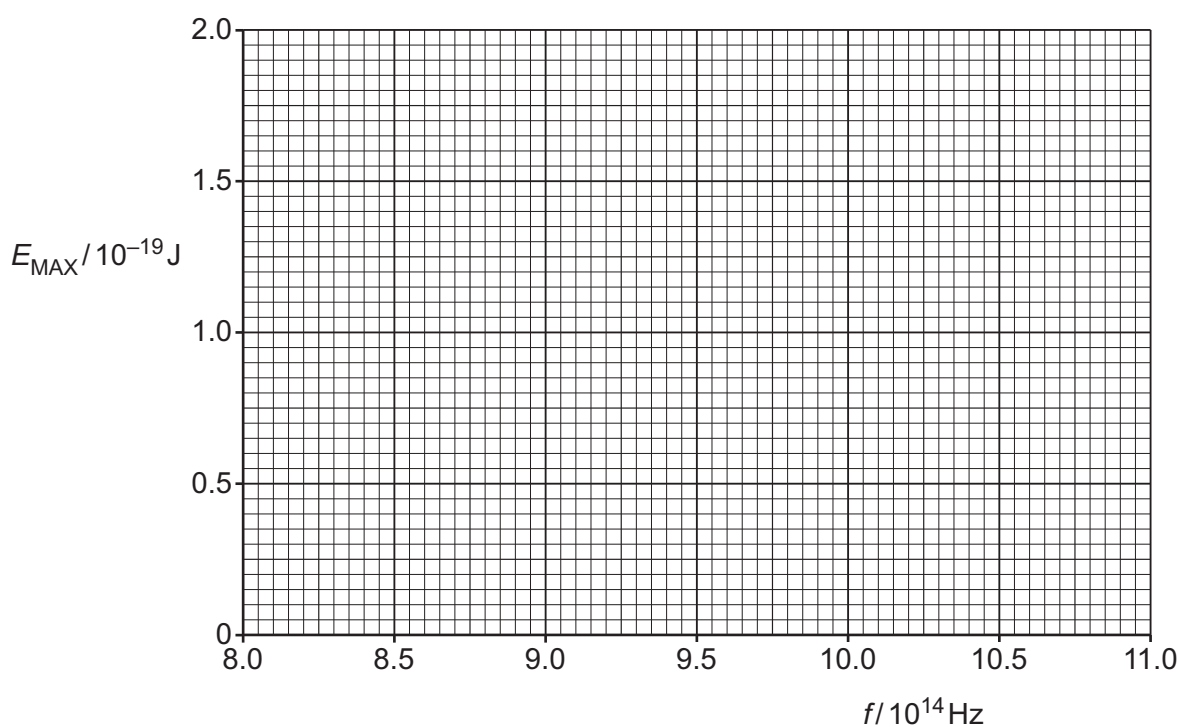


Fig. 8.1

[3]

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